ISLR v1 pg 406 - ISLRv2 pg 544

10.4 Lab1: PCA

In [1]:

```
states=row.names(USArrests)
states
```

'Alabama' · 'Alaska' · 'Arizona' · 'Arkansas' · 'California' · 'Colorado' · 'Connecticut' · 'Delaware' · 'Florida' · 'Georgia' · 'Hawaii' · 'Idaho' · 'Illinois' · 'Indiana' · 'Iowa' · 'Kansas' · 'Kentucky' · 'Louisiana' · 'Maine' · 'Maryland' · 'Massachusetts' · 'Michigan' · 'Minnesota' · 'Mississippi' · 'Missouri' · 'Montana' · 'Nebraska' · 'Nevada' · 'New Hampshire' · 'New Jersey' · 'New Mexico' · 'New York' · 'North Carolina' · 'North Dakota' · 'Ohio' · 'Oklahoma' · 'Oregon' · 'Pennsylvania' · 'Rhode Island' · 'South Carolina' · 'South Dakota' · 'Tennessee' · 'Texas' · 'Utah' · 'Vermont' · 'Virginia' · 'Washington' · 'West Virginia' · 'Wisconsin' · 'Wyoming'

In [2]:

```
names(USArrests)
```

'Murder' · 'Assault' · 'UrbanPop' · 'Rape'

In [3]:

```
apply(USArrests,2,mean)
```

Murder: 7.788 **Assault:** 170.76 **UrbanPop:** 65.54 **Rape:** 21.232

In [4]:

```
apply(USArrests,2,var)
```

Murder: 18.9704653061224 **Assault:** 6945.16571428572 **UrbanPop:** 209.518775510204 **Rape:** 87.7291591836735

In [5]:

```
pr.out = prcomp(USArrests, scale=TRUE)
```

In [6]:

```
names(pr.out)
```

'sdev' · 'rotation' · 'center' · 'scale' · 'x'

In [7]:

```
pr.out$center
```

Murder: 7.788 **Assault:** 170.76 **UrbanPop:** 65.54 **Rape:** 21.232

In [8]:

```
pr.out$scale
```

Murder: 4.35550976420929 **Assault:** 83.3376608400171 **UrbanPop:** 14.4747634008368 **Rape:** 9.36638453105965

In [9]:

```
pr.out$rotation
```

A matrix: 4 × 4 of type dbl

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	PC1	PC2	PC3	PC4
Murder	-0.5358995	-0.4181809	0.3412327	0.64922780
Assault	-0.5831836	-0.1879856	0.2681484	-0.74340748
UrbanPop	-0.2781909	0.8728062	0.3780158	0.13387773
Rape	-0.5434321	0.1673186	-0.8177779	0.08902432

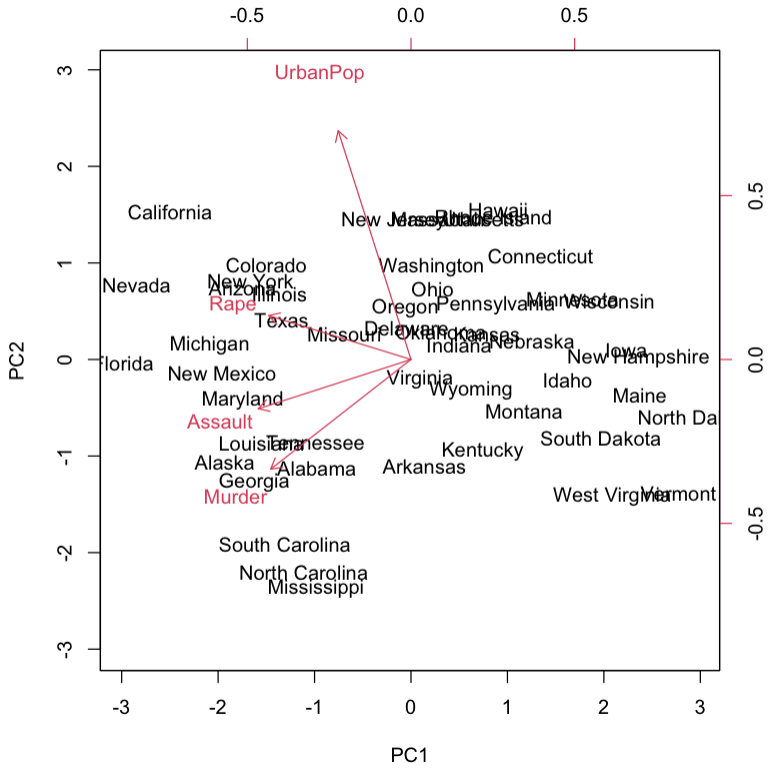
In [10]:

```
dim(pr.out$x)
```

50 · 4

In [11]:

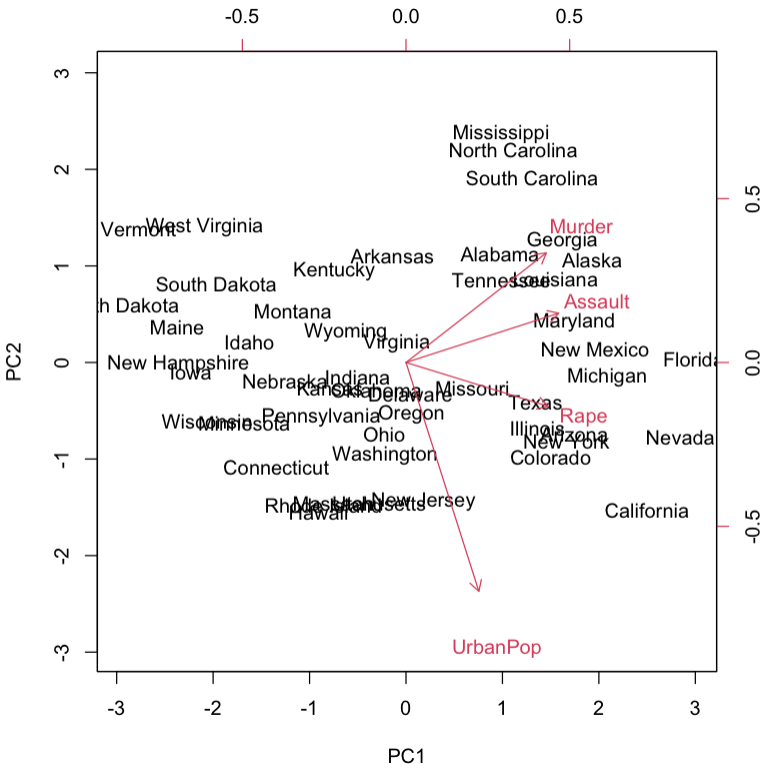
```
biplot(pr.out,scale=0)
```



In [12]:

```
pr.out$rotation=-pr.out$rotation
pr.out$x=-pr.out$x
biplot(pr.out,scale=0)
```

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In [13]:

```
pr.out$sdev
```

1.57487827439123 · 0.994869414817765 · 0.597129115502527 · 0.41644938195396

In [14]:

```
pr.var=pr.out$sdev^2
pr.var
```

2.48024157914949 · 0.989765152539842 · 0.35656318058083 · 0.173430087729835

In [15]:

```
pve = pr.var/sum(pr.var)

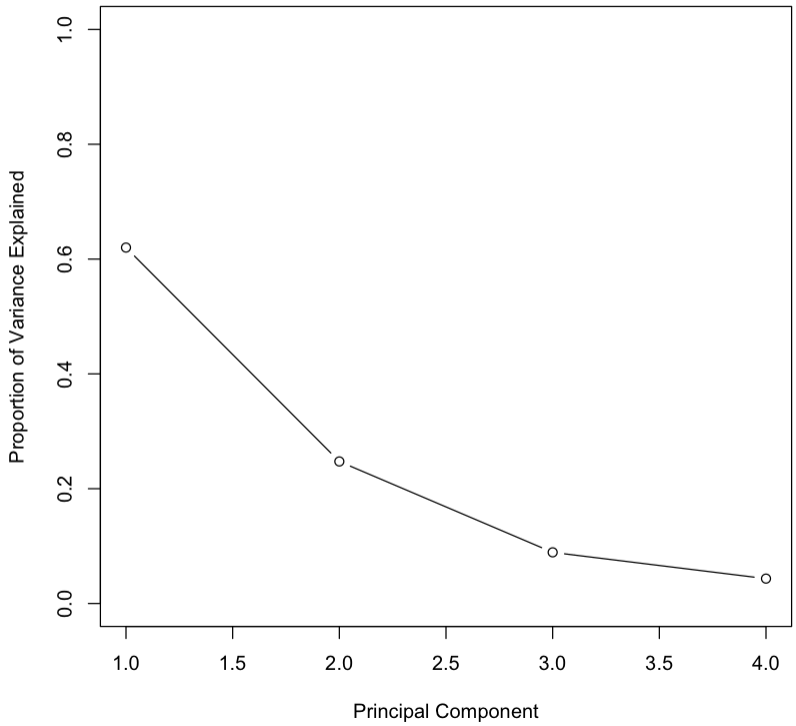
pve
```

0.620060394787373 · 0.24744128813496 · 0.0891407951452076 · 0.0433575219324588

In [16]:

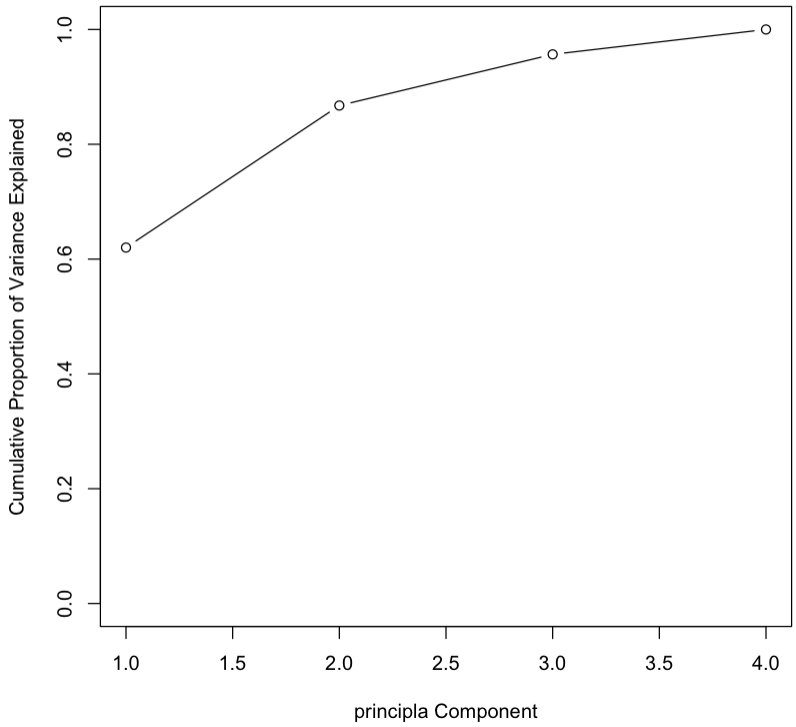
```
plot(pve, xlab="Principal Component", ylab="Proportion of Variance Explained", ylim=c(0,
```

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In [18]:

```
plot(cumsum(pve),xlab="principla Component", ylab="Cumulative Proportion of Variance Exp
```



In [19]:

```
a=c(1,2,8,-3)
cumsum(a)
```

1 · 3 · 11 · 8

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10.5 Lab 2: Clustering

In [20]:

```
set.seed(2)
x=matrix(rnorm(50*2),ncol=2)
x[1:25,1]=x[1:25,1]+3
x[1:25,2]=x[1:25,2]-4
```

In [21]:

```
km.out=kmeans(x,2,nstart=20)
```

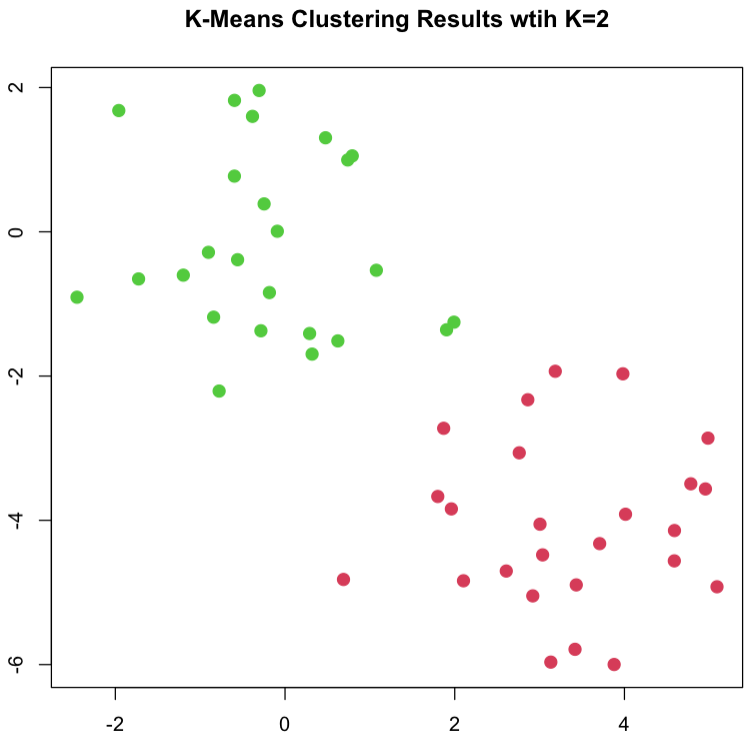
In [22]:

```
km.out$cluster
```

```
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
```

In [23]:

```
plot(x,col=(km.out$cluster+1),main="K-Means Clustering Results wtih K=2", xlab="", ylab=
```



In [24]:

```
set.seed(4)
km.out=kmeans(x,3,nstart=20)
km.out
```

```
K-means clustering with 3 clusters of sizes 17, 23, 10

Cluster means:
        [,1]        [,2]
1  3.7789567 -4.56200798
2 -0.3820397 -0.08740753
3  2.3001545 -2.69622023
```

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```
Clustering vector:
 [1] 1 3 1 3 1 1 1 3 1 3 1 3 1 3 1 3 1 1 1 1 1 3 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2
[39] 2 2 2 2 2 3 2 3 2 2 2 2

Within cluster sum of squares by cluster:
[1] 25.74089 52.67700 19.56137
 (between_SS / total_SS =  79.3 %)

Available components:

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
[6] "betweenss"    "size"         "iter"         "ifault"
```

In [26]:

```
set.seed(3)
km.out=kmeans(x,3,nstart=1)
km.out$withinss
```

25.7408883900286 · 19.5613749770553 · 52.6770041123144

In [28]:

```
km.out=kmeans(x,3,nstart=20)
km.out$tot.withinss
```

97.9792674793982

10.5.2 Hierarchial Clustering

In [29]:

```
hc.complete=hclust(dist(x),method="complete")
```

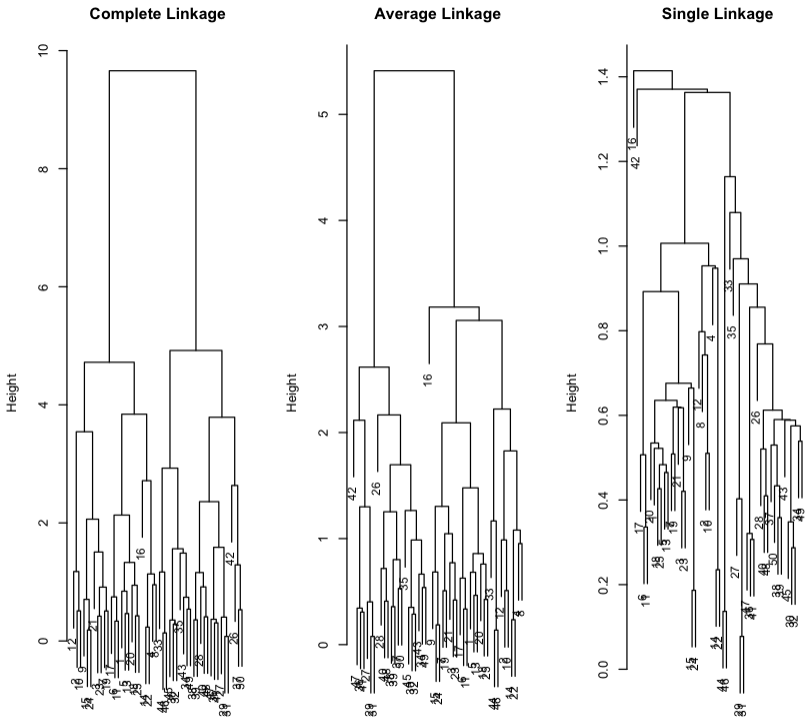
In [30]:

```
hc.average=hclust(dist(x),method="average")
hc.single=hclust(dist(x),method="single")
```

In [31]:

```
par(mfrow=c(1,3))
plot(hc.complete,main="Complete Linkage",xlab="",sub="",cex=.9)
plot(hc.average,main="Average Linkage",xlab="", sub="", cex=.9)
plot(hc.single,main="Single Linkage",xlab="",sub="",cex=.9)
```

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In [32]:

```
cutree(hc.complete,2)
cutree(hc.average,2)
cutree(hc.single,2)
```

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In [33]:

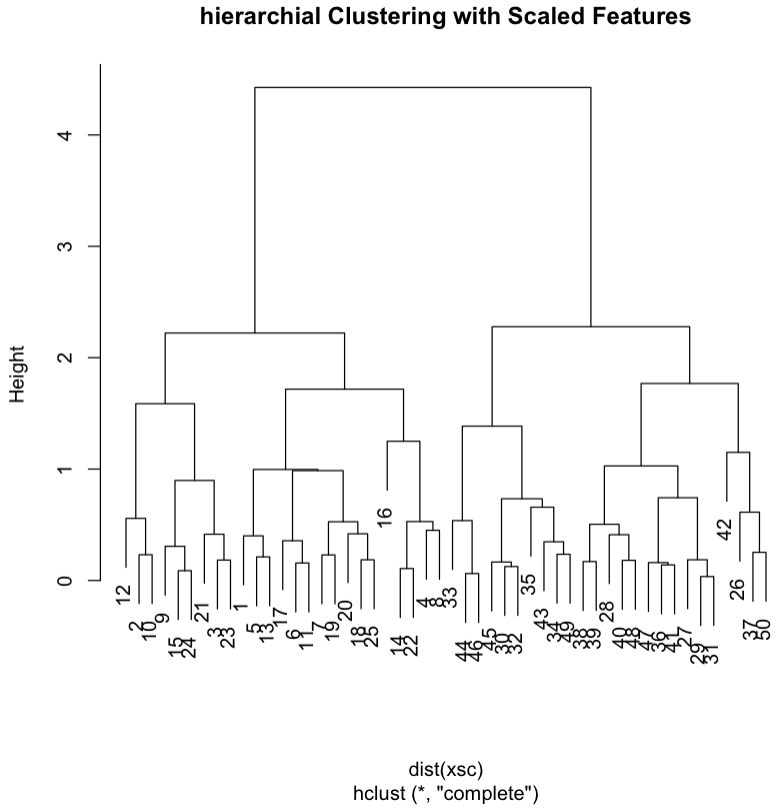
```
cutree(hc.single,4)
```

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In [34]:

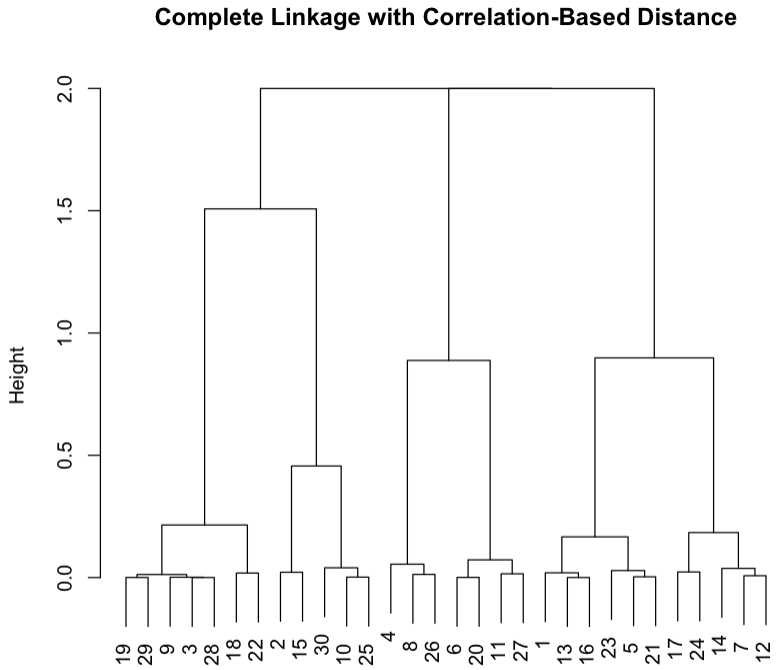
```
xsc=scale(x)
plot(hclust(dist(xsc),method="complete"),main="hierarchial Clustering with Scaled Featur
```

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In [39]:

```
x=matrix(rnorm(30*3),ncol=3)
dd=as.dist(1-cor(t(x)))
plot(hclust(dd,method="complete"),main="Complete Linkage with Correlation-Based Distance
```



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